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Al-Kufa University
Engineering College



Electronic And Communication Department

3rd stage

LABORATORY MANUAL

Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

Experiment 6

Synchronous Counters

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1. Aims:

In this experiment basic synchronous counters will be examined and the concept of decade counters will be introduced.

2. Theory :

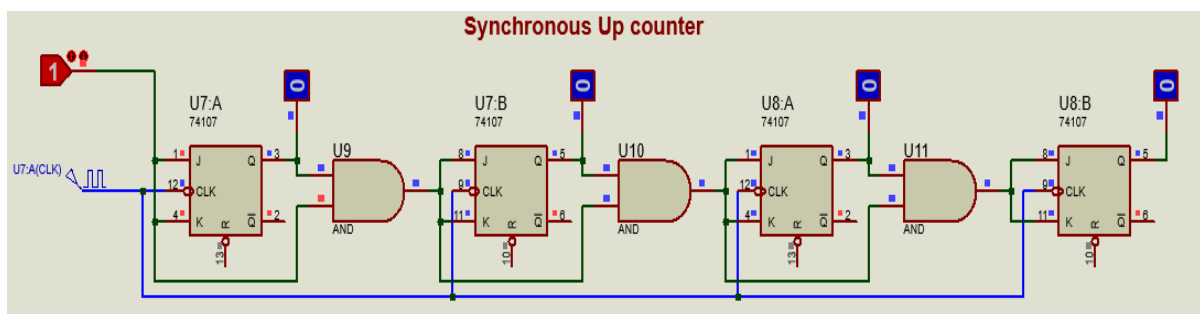
2.1 synchronous counter

Synchronous counters are different from ripple counters in that clock pulses are applied to the inputs of all flip-flops. A common clock triggers all flip-flops simultaneously, rather than one at a time in succession as in a ripple counter.

2.1.1 Up counter

To design Synchronous up counter this condition needs to be applied:

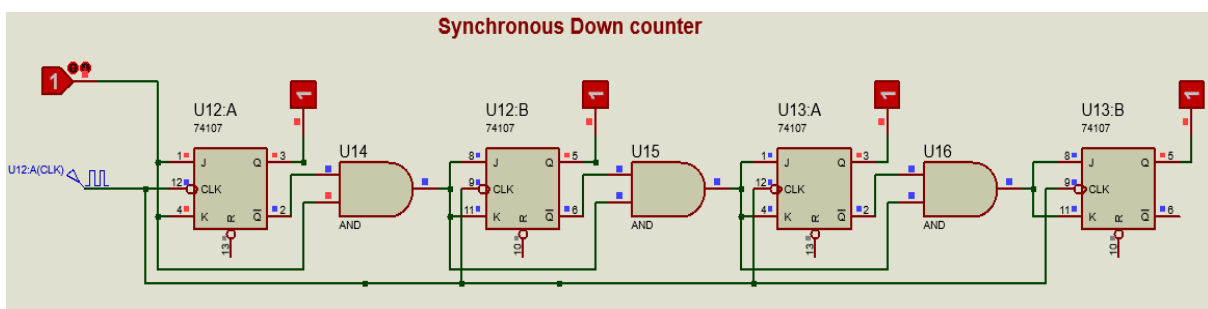
(A flip-flop in any other position is complemented when all the bits in the lower significant positions are equal to 1)



2.1.2 Down counter

To design Synchronous down counter this condition needs to be applied:

(A flip-flop in any other position is complemented when all the bits in the lower significant positions are equal to 0)



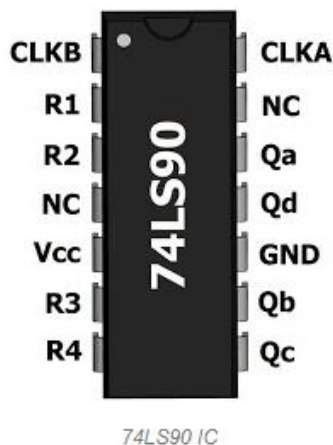
2.2 Decade Counter:

Counters can be designed to generate any desired sequence of states. A divide-by- N counter (also known as a modulo- N counter) is a counter that goes through a repeated sequence of N states. The sequence may follow the binary count or may be any other arbitrary sequence. Counters are used to generate timing signals to control the sequence of operations in a digital system. Counters can also be constructed by means of shift registers.

The term Modulus is the total number of counts that a counter has a capacity of counting pulses. When a counter counts n - counts, it reaches zero, called modulus n -counter. Examples are mod-8 counter, mod-16 counter, etc. The counting range of the n -bit modulus binary counter is from 0 to 2^n-1 .

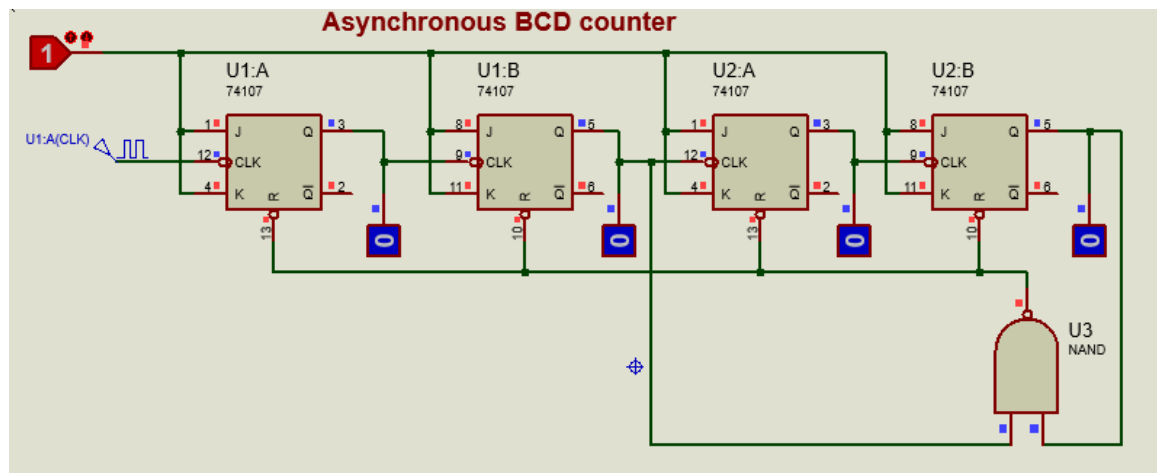
2.2.1 BCD counter

A BCD (Binary Coded Decimal) counter also termed as decade counter is a series type of digital counter which is designed to count ten digits. 74LS90 decade counter IC description and pin configuration is shown below. It is the most widely used 14-pin integrated chip.



Pin No	Function	Name
1	Clock input 2	Input2
2	Reset1	R1
3	Reset2	R2
4	Not connected	NC
5	Supply voltage; 5V (4.75V – 5.25V)	Vcc
6	Reset3	R3
7	Reset4	R4
8	Output 3, BCD Output bit 2	Q_c
9	Output 2, BCD Output bit 1	Q_b
10	Ground (0V)	Ground
11	Output 4, BCD Output bit 3	Q_d
12	Output 1, BCD Output bit 0	Q_a
13	Not connected	NC
14	Clock input 1	Input1

To make a digital counter which counts from 1 to 10, we need to have the counter count only the binary numbers 0000 to 1001. That is from 0 to 9 in decimal. Once the count value is reached, resetting them returns the counter back to zero to start again. The figure below shows asynchronous BCD counter where you can notice that it reaches back to zero when the output equals 1010.



3. Procedures:

3.1 Procedure 1:

1. construct the circuit shown in section 2.1.1, record the output indications and compute the decimal equivalent.

Synchronous Up-Counter Output Table					
INPUT	OUTPUTS				
Number of CLK I	8=L4	4=L3	2=L2	1=L1	Decimal Equivalent
0	0	0	0	0	0
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

3.2 Procedure 2 :

1. For the state table of a synchronous BCD counter listed in the Table below, use k map to derive the flip-flops equations.

Present State				Next State				Flip-Flop Inputs			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

2. From the derived equations use T flip flop to design synchronous BCD counter. Draw the circuit below



4. Discussion

- compare between synchronous and asynchronous counters. What are the advantages and limitations of each one of them?